Norris Phase Report 5

Held-out evaluation, confirmatory tests, over-editing and error screens

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Repository github.com/SHAZAN01/csudh-norris-ape-2026, branch phase4-main

Phase 5 complete. All 24 conditions scored on FLORES+ devtest under the frozen protocol; every number independently recomputed.

Executive summary

The frozen evaluator scored 24,288 post-edited outputs, the sentence-split NLLB baseline (S0) and the unsplit diagnostic on 1,012 devtest items per direction: 16,973 unique COMET-22 segments, BLEU and chrF++ with signatures, two Holm-corrected families of 24 paired-bootstrap tests, harm bands, protect-pool over-editing and preservation screens. No condition used baseline fallback, so all effects are editing behaviour.

GPT-4o Mini improved every condition: +0.26 to +1.00 COMET points, all twelve significant after Holm correction, with BLEU and chrF++ also above S0 in all twelve. Five of these gains (all Spanish-pair) have intervals within ±0.5 points: real but small. **Llama 3 8B Instruct never improved significantly**: it was significantly worse in six conditions (all four reason-then-edit directions, and zero/one-shot En→Tl), its other six intervals lie within ±0.5 points, and its BLEU and chrF++ fall below S0 in all twelve.

Over-editing. Of 1,179 strong drafts (S0 COMET ≥ 0.90), Llama over-edited 132 (zero-shot, 11.2%), 133 (one-shot, 11.3%) and 247 (reason-then-edit, 20.9%); GPT-4o Mini over-edited 10, 51 and 25 (0.8%, 4.3%, 2.1%). The ordering holds at all three sensitivity thresholds. **Strategies:** 15 of 24 contrasts are significant; retrieved one-shot is the best GPT-4o Mini strategy, and reason-then-edit was never significantly better than a strategy without diagnosis.

Verification	Result
Frozen code	Evaluator and statistics module hashes equal the freeze; evaluation bound to freeze 6e6259a9…
Coverage	24 conditions × 1,012 IDs; all records validated against Phase 4 run hashes
Independent recomputation (F1, F2)	All 48 deltas, intervals, raw and Holm p-values recomputed from 24,288 raw segment scores: maximum difference 0
Counts and pools	Bands, edits, protect pools and over-edit numerators recomputed for all 24 conditions: 0 mismatches
Generated tables and macros	4 files byte-identical to regeneration from main_results.json
Manuscript audit	61 held-out claims plus generated files verified; five planted errors all detected
Statistical procedure	Passed 8 synthetic fixtures (identity, constant shift, symmetry, null calibration, Holm) before scoring

1. Systems on devtest

System	En→Tl C / BLEU / chrF++	Tl→En C / BLEU / chrF++	En→Es C / BLEU / chrF++	Es→En C / BLEU / chrF++
S0 split NLLB	85.20 / 36.2 / 61.1	87.57 / 49.2 / 68.5	86.67 / 28.8 / 54.6	87.37 / 32.1 / 59.4
Unsplit NLLB (diagnostic)	84.81 / 35.8 / 60.4	87.49 / 49.0 / 68.2	86.46 / 28.6 / 54.1	87.32 / 32.0 / 59.0
GPT-4o Mini Zero	85.84 / 37.2 / 61.8	88.39 / 50.1 / 69.7	86.95 / 29.1 / 54.8	87.71 / 32.7 / 59.9
GPT-4o Mini One	86.15 / 38.0 / 62.4	88.57 / 49.9 / 70.1	87.20 / 29.5 / 55.1	87.73 / 32.3 / 59.8
GPT-4o Mini Reason	85.71 / 36.7 / 61.5	88.22 / 50.0 / 69.3	86.93 / 28.9 / 54.6	87.64 / 32.4 / 59.7
Llama 3 8B Zero	84.72 / 33.9 / 59.0	87.47 / 47.3 / 67.6	86.68 / 28.0 / 54.0	87.36 / 31.5 / 59.0
Llama 3 8B One	84.81 / 34.0 / 59.6	87.60 / 46.9 / 67.3	86.85 / 28.1 / 54.1	87.33 / 30.9 / 58.6
Llama 3 8B Reason	83.76 / 33.5 / 58.6	86.25 / 45.5 / 65.6	86.16 / 27.7 / 53.6	86.63 / 30.7 / 58.1

COMET-22 ×100 (Unbabel/wmt22-comet-da @ 2760a223); BLEU nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.6.0; chrF++ nrefs:1|case:mixed|eff:yes|nc:6|nw:2|space:no|version:2.6.0 (word order 2, as frozen; the roadmap's nw:1 is a documented deviation).

2. Primary family F1: each condition minus S0 (Holm over 24)

Editor	Strategy	En→Tl	Tl→En	En→Es	Es→En
GPT-4o Mini	Zero-shot	+0.64* [+0.50, +0.77]	+0.83* [+0.69, +0.97]	+0.28* [+0.17, +0.39] ‡	+0.34* [+0.24, +0.46] ‡
GPT-4o Mini	One-shot	+0.95* [+0.78, +1.12]	+1.00* [+0.83, +1.18]	+0.53* [+0.37, +0.69]	+0.36* [+0.24, +0.50] ‡
GPT-4o Mini	Reason-then-edit	+0.51* [+0.38, +0.64]	+0.65* [+0.50, +0.81]	+0.26* [+0.14, +0.38] ‡	+0.27* [+0.17, +0.39] ‡
Llama 3 8B	Zero-shot	-0.48* [-0.65, -0.32]	-0.10 [-0.26, +0.05] ‡	+0.00 [-0.18, +0.19] ‡	-0.01 [-0.13, +0.12] ‡
Llama 3 8B	One-shot	-0.39* [-0.57, -0.21]	+0.03 [-0.12, +0.18] ‡	+0.18 [-0.05, +0.40] ‡	-0.04 [-0.24, +0.16] ‡
Llama 3 8B	Reason-then-edit	-1.44* [-1.70, -1.21]	-1.32* [-1.57, -1.07]	-0.51* [-0.71, -0.31]	-0.74* [-0.93, -0.55]

Δ in COMET points with paired percentile 95% CI (10,000 resamples, seed 42). * significant after Holm (all marked p_Holm = 0.0024, the minimum attainable with B = 10,000 and 24 tests). ‡ interval within the ±0.5-point practical band (flag only, not an equivalence test). Unmarked differences are inconclusive.

3. Strategy family F2 (Holm over 24)

Editor	Contrast	En→Tl	Tl→En	En→Es	Es→En
GPT-4o Mini	One – Zero	+0.31* [+0.19, +0.43] p_H=0.002	+0.17* [+0.08, +0.27] p_H=0.002	+0.24* [+0.13, +0.36] p_H=0.002	+0.02 [-0.05, +0.09] p_H=1.000
GPT-4o Mini	Reason – Zero	-0.13 [-0.23, -0.03] p_H=0.102	-0.18* [-0.27, -0.08] p_H=0.002	-0.03 [-0.12, +0.07] p_H=1.000	-0.07 [-0.13, -0.01] p_H=0.246
GPT-4o Mini	Reason – One	-0.44* [-0.58, -0.30] p_H=0.002	-0.35* [-0.47, -0.22] p_H=0.002	-0.27* [-0.40, -0.13] p_H=0.002	-0.09 [-0.18, +0.00] p_H=0.323
Llama 3 8B	One – Zero	+0.10 [-0.09, +0.28] p_H=1.000	+0.13 [+0.01, +0.26] p_H=0.246	+0.17 [-0.02, +0.35] p_H=0.374	-0.03 [-0.20, +0.12] p_H=1.000
Llama 3 8B	Reason – Zero	-0.96* [-1.22, -0.72] p_H=0.002	-1.22* [-1.47, -0.97] p_H=0.002	-0.51* [-0.73, -0.31] p_H=0.002	-0.73* [-0.89, -0.57] p_H=0.002
Llama 3 8B	Reason – One	-1.06* [-1.34, -0.79] p_H=0.002	-1.35* [-1.61, -1.10] p_H=0.002	-0.68* [-0.91, -0.44] p_H=0.002	-0.70* [-0.91, -0.48] p_H=0.002

GPT-4o Mini: one-shot beats zero-shot in En→Tl, Tl→En and En→Es (Es→En inconclusive); reason-then-edit is below one-shot in the same three and below zero-shot only in Tl→En. Llama: one-shot and zero-shot never differ significantly; reason-then-edit is significantly worse than both in all four directions. All 16 reason-then-edit point estimates are negative.

4. Segmentation diagnostic (not in a confirmatory family)

S0 split minus unsplit NLLB: En→Tl +0.39 [+0.25, +0.57]; Tl→En +0.08 [+0.02, +0.15]; En→Es +0.21 [+0.12, +0.31]; Es→En +0.05 [-0.04, +0.13]. Splitting helps both out-of-English directions and Tl→En; Es→En is inconclusive. On dev, Tl→En had been inconclusive.

5. Harm bands and over-editing (RQ3; 5.txt bins)

Editor	Strategy	Dir.	Edited %	Improved %	Unchanged %	Harmed % (roadmap over-edit rate)	Protect over-edits .01 / .02 / .05
GPT-4o Mini	Zero-shot	En→Tl	46.5	12.7	85.5	1.8	7 / 3 / 1 of 234
GPT-4o Mini	Zero-shot	Tl→En	40.7	14.3	84.8	0.9	9 / 3 / 1 of 377
GPT-4o Mini	Zero-shot	En→Es	22.8	5.2	93.5	1.3	4 / 3 / 2 of 275
GPT-4o Mini	Zero-shot	Es→En	26.8	5.5	93.6	0.9	2 / 1 / 1 of 293
GPT-4o Mini	One-shot	En→Tl	89.6	20.8	73.6	5.5	17 / 6 / 2 of 234
GPT-4o Mini	One-shot	Tl→En	85.4	21.0	73.9	5.0	54 / 26 / 3 of 377
GPT-4o Mini	One-shot	En→Es	68.5	12.0	84.2	3.9	27 / 7 / 1 of 275
GPT-4o Mini	One-shot	Es→En	77.6	8.2	88.2	3.6	42 / 12 / 2 of 293
GPT-4o Mini	Reason-then-edit	En→Tl	41.7	11.9	85.0	3.2	7 / 5 / 2 of 234
GPT-4o Mini	Reason-then-edit	Tl→En	37.8	13.9	82.7	3.4	21 / 13 / 3 of 377
GPT-4o Mini	Reason-then-edit	En→Es	16.2	4.5	94.1	1.4	3 / 3 / 0 of 275
GPT-4o Mini	Reason-then-edit	Es→En	19.1	4.8	93.8	1.4	5 / 4 / 1 of 293
Llama 3 8B	Zero-shot	En→Tl	81.1	9.4	72.2	18.4	69 / 42 / 11 of 234
Llama 3 8B	Zero-shot	Tl→En	53.5	7.3	84.8	7.9	56 / 35 / 11 of 377
Llama 3 8B	Zero-shot	En→Es	70.7	8.6	81.9	9.5	51 / 29 / 11 of 275
Llama 3 8B	Zero-shot	Es→En	57.7	4.9	88.4	6.6	44 / 26 / 3 of 293
Llama 3 8B	One-shot	En→Tl	88.4	11.3	72.3	16.4	63 / 35 / 12 of 234
Llama 3 8B	One-shot	Tl→En	71.3	10.2	81.6	8.2	67 / 37 / 8 of 377
Llama 3 8B	One-shot	En→Es	85.5	13.9	76.6	9.5	57 / 28 / 4 of 275
Llama 3 8B	One-shot	Es→En	85.9	9.0	82.5	8.5	72 / 33 / 8 of 293
Llama 3 8B	Reason-then-edit	En→Tl	92.0	9.0	61.7	29.3	94 / 57 / 22 of 234
Llama 3 8B	Reason-then-edit	Tl→En	81.4	7.8	67.4	24.8	149 / 107 / 44 of 377
Llama 3 8B	Reason-then-edit	En→Es	57.6	6.2	80.3	13.4	59 / 32 / 13 of 275
Llama 3 8B	Reason-then-edit	Es→En	69.6	6.1	76.3	17.6	89 / 51 / 15 of 293

Bins on native ΔCOMET: improved > +0.02, unchanged within ±0.02, harmed < –0.02 (the roadmap's "over-editing rate"). The frozen primary over-editing measure is stricter: a strong draft (S0 ≥ 0.90) whose text changed and lost > 0.02, over all protect items; 0.01 and 0.05 are sensitivity thresholds. Protect pools: En→Tl 234, Tl→En 377, En→Es 275, Es→En 293.

6. Gains and losses by baseline quality (RQ1)

Editor	Strategy	En→Tl weak / mid / strong	Tl→En weak / mid / strong	En→Es weak / mid / strong	Es→En weak / mid / strong
GPT-4o Mini	Zero-shot	+2.04 / +0.44 / +0.03	+4.21 / +0.85 / +0.06	+1.63 / +0.23 / -0.12	+3.20 / +0.21 / +0.06
GPT-4o Mini	One-shot	+3.11 / +0.60 / +0.15	+5.29 / +1.11 / -0.10	+3.29 / +0.37 / -0.16	+4.32 / +0.23 / -0.15
GPT-4o Mini	Reason-then-edit	+1.93 / +0.27 / -0.01	+4.04 / +0.64 / -0.07	+1.91 / +0.11 / -0.03	+3.27 / +0.14 / -0.03
Llama 3 8B	Zero-shot	+0.68 / -0.68 / -0.91	+1.65 / -0.07 / -0.52	+2.41 / -0.11 / -0.63	+2.75 / -0.11 / -0.33
Llama 3 8B	One-shot	+0.60 / -0.49 / -0.90	+1.89 / +0.09 / -0.45	+2.75 / +0.03 / -0.46	+3.82 / -0.17 / -0.52
Llama 3 8B	Reason-then-edit	-0.51 / -1.62 / -1.74	+1.05 / -1.14 / -2.09	+1.78 / -0.76 / -0.80	+1.92 / -0.91 / -0.88

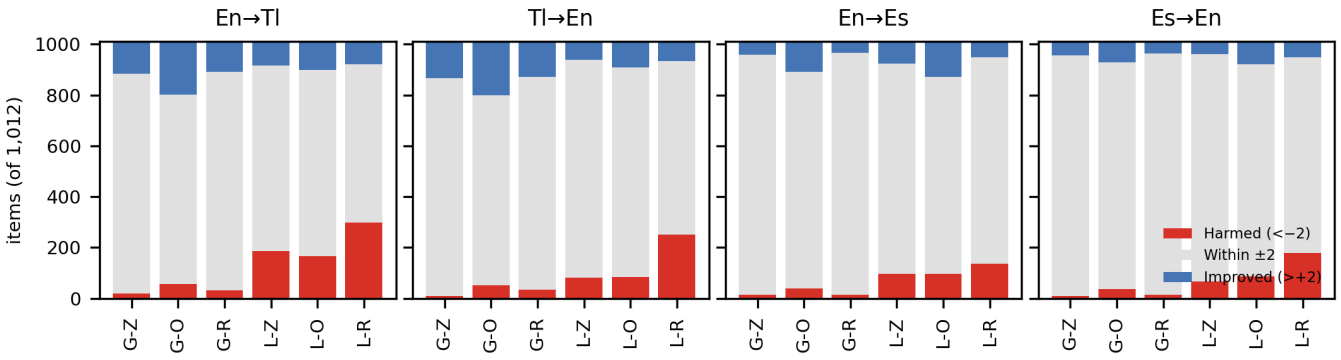
Mean ΔCOMET points by S0 quality: weak < 0.80 (En→Tl 183, Tl→En 82, En→Es 104, Es→En 60 items), middle, strong ≥ 0.90. Editing helps weak drafts in 23 of 24 conditions; on strong drafts GPT-4o Mini stays within ±0.16 points while Llama loses 0.33–2.09 points. The Tagalog pair's larger GPT-4o Mini gains coincide with its larger share of weak drafts; within the weak pool, GPT-4o Mini one-shot gains are similar across pairs (+3.11 to +5.29). Descriptive; no confirmatory test of pair or direction differences was planned.

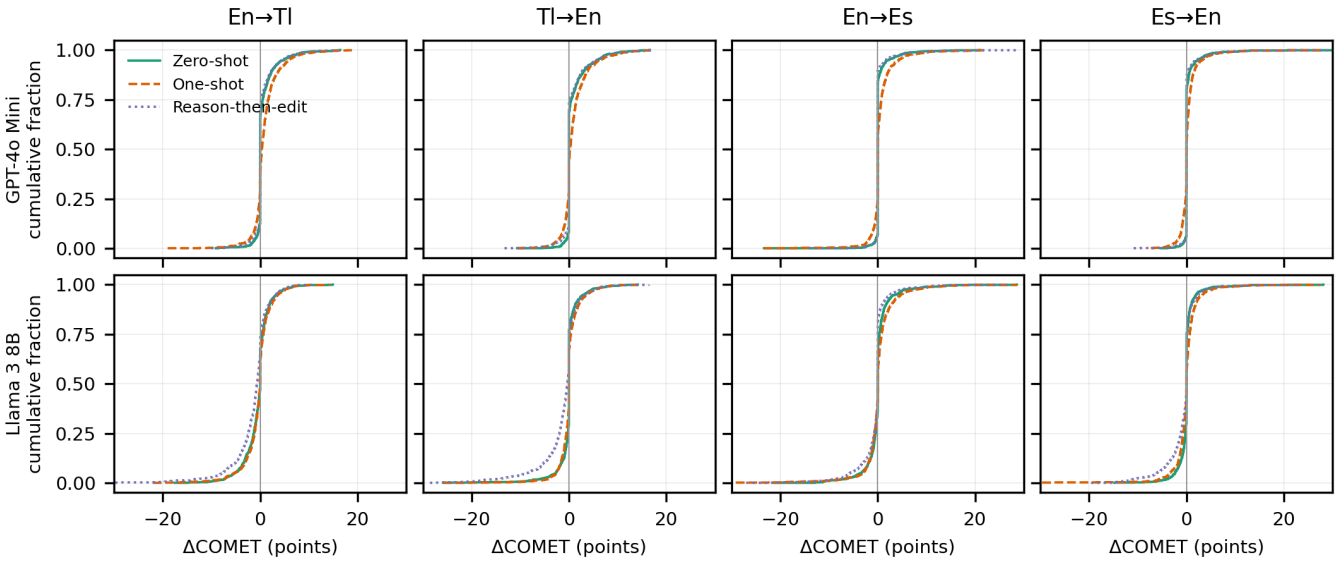
7. Programmatic error screens (5.txt taxonomy)

Editor	Strategy	Dir.	Digit change / loss	Capitalized token lost	Language drift (S0)	Length anomaly	Sentence drop	Mean norm. edit dist.
GPT-4o Mini	Zero-shot	En→Tl	5 / 5	11	3 (3)	0	1	0.058
GPT-4o Mini	Zero-shot	Tl→En	5 / 5	5	2 (1)	0	1	0.094
GPT-4o Mini	Zero-shot	En→Es	7 / 4	0	1 (1)	0	0	0.044
GPT-4o Mini	Zero-shot	Es→En	7 / 7	9	1 (1)	0	4	0.058
GPT-4o Mini	One-shot	En→Tl	8 / 6	21	2 (3)	0	1	0.078
GPT-4o Mini	One-shot	Tl→En	9 / 7	8	2 (1)	0	1	0.111
GPT-4o Mini	One-shot	En→Es	15 / 7	1	1 (1)	0	3	0.065
GPT-4o Mini	One-shot	Es→En	10 / 9	15	1 (1)	0	4	0.091
GPT-4o Mini	Reason-then-edit	En→Tl	5 / 4	5	3 (3)	0	1	0.059
GPT-4o Mini	Reason-then-edit	Tl→En	6 / 6	6	2 (1)	0	1	0.092
GPT-4o Mini	Reason-then-edit	En→Es	4 / 4	3	1 (1)	0	1	0.061
GPT-4o Mini	Reason-then-edit	Es→En	4 / 4	8	1 (1)	0	4	0.062
Llama 3 8B	Zero-shot	En→Tl	23 / 7	11	5 (3)	0	1	0.110
Llama 3 8B	Zero-shot	Tl→En	12 / 9	16	2 (1)	0	1	0.101
Llama 3 8B	Zero-shot	En→Es	20 / 13	12	5 (1)	0	1	0.097
Llama 3 8B	Zero-shot	Es→En	10 / 10	16	1 (1)	0	4	0.098
Llama 3 8B	One-shot	En→Tl	17 / 8	27	2 (3)	1	1	0.107
Llama 3 8B	One-shot	Tl→En	10 / 7	16	1 (1)	0	1	0.112
Llama 3 8B	One-shot	En→Es	22 / 15	7	2 (1)	3	2	0.121
Llama 3 8B	One-shot	Es→En	15 / 13	18	1 (1)	1	5	0.135
Llama 3 8B	Reason-then-edit	En→Tl	10 / 7	6	6 (3)	1	1	0.088
Llama 3 8B	Reason-then-edit	Tl→En	8 / 7	6	10 (1)	0	1	0.108
Llama 3 8B	Reason-then-edit	En→Es	13 / 9	4	2 (1)	0	1	0.075
Llama 3 8B	Reason-then-edit	Es→En	12 / 12	10	5 (1)	0	4	0.101

Counts out of 1,012. Number/entity corruption: digit multiset changed or baseline digit lost, and capitalized source tokens newly absent. Length anomaly: output/source word ratio > 2.5 or < 0.4. Language drift: langdetect 1.0.9 (seed 0; replaces the roadmap's fastText), with the S0 rate in brackets. Sentence drop: fewer target sentences than S0 on multi-sentence items. Edit distance: character Levenshtein over edited items, normalized by length. These are screens, not gold error labels; legitimate localisation can trigger them.

8. Figures





Top: improved / within band / harmed counts per condition. Bottom: empirical CDFs of segment ΔCOMET per direction (x-axis clipped at ±30 points).

9. Answers to the research questions (from frozen analyses)

RQ1 (pairs, conditional on quality). GPT-4o Mini gains are larger for Tagalog (+0.51 to +1.00) than Spanish (+0.26 to +0.53); Llama's losses are also largest for Tagalog. The difference tracks the share of weak drafts more than language: within the weak pool, gains are similar across pairs. Resource level is confounded with language, direction, domain and baseline quality; no causal resource claim is made.

RQ2 (strategies). Retrieved one-shot is the strongest GPT-4o Mini strategy (significant over zero-shot in three directions). Reason-then-edit never helps: it is significantly worse than one-shot for GPT-4o Mini in three directions and than both other strategies for Llama in all four.

RQ3 (over-editing). Editors change strong drafts often; GPT-4o Mini rarely harms them (≤ 4.3%), whereas Llama harms 11–21%, worst with reason-then-edit (107 of 377 strong Tl→En drafts). Edit rate alone does not predict harm: GPT-4o Mini one-shot edits 80% of drafts yet over-edits only 4.3% of strong drafts, while Llama zero-shot edits fewer (66%) and over-edits 11.2%.

RQ4 (direction). Direction effects are small and inconsistent: GPT-4o Mini gains slightly more into English in the Tagalog pair for every strategy, but not in the Spanish pair. FLORES+ sources are largely English-original, so reverse directions carry translationese; no direction claim beyond these estimates is made.

10. Source-based case review (remaining Phase 5 item)

The approved plan calls for source review of flagged held-out cases. Text is not in the numeric evidence, so a deterministic, rule-based extraction was added (src/analysis/extract_review_cases.py): per condition, the strong draft with the largest over-edit loss, the largest improvement, the first digit-loss flag and the first language-drift flag. Up to 96 cases go to a private file. Their review completes this report and supplies two or three illustrations for the manuscript's Discussion. Rules were fixed before any held-out text was read.

11. Deviations, incidents and integrity

• No protocol deviation after the freeze. Evaluator, statistics, thresholds and families exactly as frozen; no test added to F1 or F2 after viewing devtest.
• Figure script (not part of the frozen analysis) failed under Colab's inline matplotlib backend; fixed by forcing the Agg backend. No numbers affected.
• Roadmap differences documented at the freeze and unchanged: chrF++ word order 2; langdetect instead of fastText; reason-then-edit instead of "contrastive CoT"; harm and protect-pool over-editing reported separately.
• The S0 dev-reproduction disposition (Report 3.2, Section 5) still awaits PI acknowledgement; S0 is identical for all conditions, so comparisons are unaffected.

12. Manuscript integration and next steps

The ICNLP manuscript now reports the held-out experiment: abstract, results (Table I generated, over-editing table, band figure), discussion and conclusion, at exactly 5 pages including references. Every held-out number in the prose is registered and audited against these results; development findings remain labelled as development evidence. Two markers remain: the source-reviewed cases (Section 10) and venue-required AI-assistance and funding statements, which need PI confirmation. Then: internal audit and reviewer memo (7.txt), reproducibility packaging (8.txt), complete draft to the PI by 24 November.

Evidence

Phase 5 evidence ZIP 02ca2839350caeee78b07071d2e913864f4a753fe3e66d77f84b9c4edb088da3
Evaluation aaff72ad3e0a432c7b12c341abe01f6439e37f12be5daf94579839a972e4e147
Freeze 6e6259a9cf1c7e2aca978a770fa85d86596f31b0e8b3c55f42635203e4f7462c